

Project 1: Statistical study of Atlanta monthly temperatures

Descriptive statistics, normality, inference

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2025–2026

Context

The dataset `AvTempAtlanta.txt` contains the monthly average temperatures (degrees Fahrenheit) of Atlanta, USA, over several decades. The goal of this project is to apply the basic statistical tools of Chapter 1 to a real dataset: descriptive statistics, normality diagnostic, confidence intervals, and hypothesis testing.

Deliverables

- A reproducible Python script (`project1_<name>.py`) with seed 42.
- A 4–6 page PDF report including all figures, tables and discussion.
- Optional 5-minute oral defence per student.

Exercise 1 – Descriptive statistics

(1) Load the data, plot the time series. Comment on the qualitative features (trend, seasonality, range).

(2) Compute and report:

- sample mean, variance, standard deviation, min, max, median;
- sample skewness and kurtosis.

Plot a histogram with kernel density estimate overlay.

Exercise 2 – Normality diagnostic

(1) Plot a Q–Q plot (`scipy.stats.probplot`) against the standard normal.

(2) Run two formal normality tests:

- Shapiro–Wilk (`scipy.stats.shapiro`);
- Jarque–Bera (`scipy.stats.jarque_bera`).

Report the test statistics and p -values. Conclude on the normality assumption at the 5% level.

Exercise 3 – Confidence interval and hypothesis test

(1) Compute the 90%, 95% and 99% t -confidence intervals for the mean monthly temperature.

(2) Test $H_0 : \mu = 60$ vs. $H_1 : \mu \neq 60$ at level 5%. Report T , the critical value, the p -value, and conclude.

(3) Repeat with $H_0 : \mu = 65$. Discuss the role of μ_0 in the test.

Exercise 4 – Yearly aggregation

(1) Aggregate the monthly data into yearly averages (12-month mean). How many yearly observations do you obtain?

(2) Run a t -test on the yearly averages: $H_0 : \mu = 60$ vs. $H_1 : \mu \neq 60$. How does the conclusion compare with the monthly version?

(3) Why does aggregation change the test sensitivity? Discuss in terms of the variance of the sample mean.

Exercise 5 – Bonus: bootstrap and assumption robustness

(1) Build a bootstrap CI for the mean ($B = 10^4$). Compare with the t -CI.

(2) If the data are not normal (cf. Exercise 2), is the t -CI still valid? Justify (CLT, sample size).

Marking grid (out of 100)

Criterion	Points
Loading + descriptive statistics + plots	15
Normality diagnostic + interpretation	15
t -CI (3 levels) + t -test ($\mu_0 \in \{60, 65\}$)	25
Yearly aggregation + comparison	15
Bootstrap (bonus) + robustness discussion	10
Reproducibility (seed, code structure, comments)	10
Report quality (figures, tables, discussion)	10
Total	100

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